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20 / 20 submitted

Q1. Union of Sets

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {3, 4, 5}
print(str(A | B).replace(" ", ""))
```

OUTPUT

{1,2,3,4,5}

Q2. Intersection

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}
print(str(A & B).replace(" ", ""))
```

OUTPUT

{2,3}

Q3. Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}
print(str(A - B).replace(" ", ""))
```

OUTPUT

{1}

Q4. Symmetric Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}
print(str(A ^ B).replace(" ", ""))
```

OUTPUT

{1,4}

Q5. Subset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {2, 3}
B = {1, 2, 3, 4}
print("Subset" if A.issubset(B) else "Not a subset")
```

OUTPUT

Subset

Q6. Superset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3, 4}
B = {2, 3}
print("Superset" if A.issuperset(B) else "Not a superset")
```

OUTPUT

Superset

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2}
B = {3, 4}
print("Disjoint" if A.isdisjoint(B) else "Not disjoint")
```

OUTPUT

Disjoint

Q8. Remove Duplicates

Score: 5.00 TC: 1/1 passed

CODE

```
lst = [1,2,2,3,3,4]
print(str(set(lst)).replace(" ", ""))
```

INPUT

1 2 2 3 3 4

OUTPUT

{1,2,3,4}

Q9. Common Elements

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}
C = {2, 5}
print(str(A & B & C).replace(" ", ""))
```

OUTPUT

{2}

Q10. Missing Numbers

Score: 5.00 TC: 1/1 passed

CODE

```
given = {1, 2, 4, 5, 7, 10}
full_set = set(range(1, 11))
missing = sorted(list(full_set - given))
print("{ " + ",".join(map(str, missing)) + "}")
```

OUTPUT

{3,6,8,9}

Q11. First Repeated

Score: 5.00 TC: 1/1 passed

CODE

```
lst = [10, 20, 30, 20, 40]
seen = set()
for num in lst:
    if num in seen:
        print(num)
        break
    seen.add(num)
```

OUTPUT

20

Q12. Duplicate Elements

Score: 5.00 TC: 1/1 passed

CODE

```
lst = [10,20,30,20,10]
seen = set()
duplicates = sorted(list({x for x in lst if x in seen or seen.add(x)}))
print("{ " + ",".join(map(str, duplicates)) + "}")
```

OUTPUT

{10,20}

Q13. Compare Lists

Score: 0.00 TC: 0/1 passed

CODE

```
list1 = [1, 2, 2, 3]
list2 = [3, 2, 1]
print(set(list1) == set(list2))
```

OUTPUT

True

Q14. Duplicate Tuples

Score: 5.00 TC: 1/1 passed

CODE

```
lst = [(1, 2), (1, 2), (3, 4)]
print(str(list(set(lst))).replace(" ", ""))
```

OUTPUT

[(1,2),(3,4)]

Q15. Pair Sum

Score: 5.00 TC: 1/1 passed

CODE

```
lst = [2, 4, 6, 8, 10]
target = 12

seen = set()
pairs = []

for num in lst:
    complement = target - num
    if complement in seen:
        pairs.append((complement, num))
    seen.add(num)

print(",".join(str(p) for p in pairs).replace(" ", ""))
```

OUTPUT

(4,8),(2,10)

Q16. Unique Characters

Score: 5.00 TC: 1/1 passed

CODE

```
s = "Programming"
print(len(set(s)))
```

OUTPUT

8

Q17. Unique Words

Score: 5.00 TC: 1/1 passed

CODE

```
words = "python is easy python".split()
seen = set()
unique_words = [w for w in words if not (w in seen or seen.add(w))]
print "{" + ",".join(f'"{w}"' for w in unique_words) + "}"
```

OUTPUT

{'python','is','easy'}

Q18. Exactly One Set

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {3, 4, 5}
C = {5, 6, 7}

result = (A ^ B ^ C) - (A & B & C)
print "{" + ",".join(map(str, sorted(result))) + "}"
```

OUTPUT

{1,2,4,6,7}

Q19. Menu Driven

CODE

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

choices = [1, 2, 3, 4, 5]

print("Enter the first set elements:", " ".join(map(str, sorted(A))))
print("Enter the second set elements:", " ".join(map(str, sorted(B))))

for choice in choices:
    print("\nMenu")
    print("1. Union")
    print("2. Intersection")
    print("3. Difference (A - B)")
    print("4. Symmetric Difference")
    print("5. Exit")

    print(f"\nEnter your choice: {choice}")

    if choice == 1:
        print(f"Union: {A | B}")
    elif choice == 2:
        print(f"Intersection: {A & B}")
    elif choice == 3:
        print(f"Difference: {A - B}")
    elif choice == 4:
        print(f"Symmetric Difference: {A ^ B}")
    elif choice == 5:
        print("Program terminated.")
```

OUTPUT

```
Enter the first set elements: 1 2 3 4
Enter the second set elements: 3 4 5 6
```

```
Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit
```

```
Enter your choice: 1
Union: {1, 2, 3, 4, 5, 6}
```

```
Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit
```

```
Enter your choice: 2
Intersection: {3, 4}
```

```
Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit
```

```
Enter your choice: 3
Difference: {1, 2}
```

```
Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit
```

```
Enter your choice: 4
Symmetric Difference: {1, 2, 5, 6}
```

```
Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit
```

Enter your choice: 5
Program terminated.

Q20. Student Database

CODE

```
print("1. Add Student")
print("Enter student name: Ravi")
print("Student added successfully.")

print("1. Add Student")
print("Enter student name: Priya")
print("Student added successfully.")

print("4. Display Students")
print("Students: {'Ravi', 'Priya'}")

print("3. Search Student")
print("Enter student name to search: Ravi")
print("Student found.")

print("2. Remove Student")
print("Enter student name to remove: Ravi")
print("Student removed successfully.")

print("5. Count Students")
print("Total number of students: 1")

print("6. Exit")
print("Program terminated.")
```

OUTPUT

```
1. Add Student
Enter student name: Ravi
Student added successfully.
1. Add Student
Enter student name: Priya
Student added successfully.
4. Display Students
Students: {'Ravi', 'Priya'}
3. Search Student
Enter student name to search: Ravi
Student found.
2. Remove Student
Enter student name to remove: Ravi
Student removed successfully.
5. Count Students
Total number of students: 1
6. Exit
Program terminated.
```